Customer Retention Intelligence Dashboard

MIT 8339 Business Intelligence and Decision Support Systems · BankChurners dataset, 10,127 credit-card customers · Prepared for the Chief Data Officer

CARD CATEGORY: All | INCOME SEGMENT: All | AGE BAND: All | GENDER: All | RISK BAND: All

10,127 of 10,127 customers in view | Reset filters

1 · Executive overview | 2 · Customer retention analysis

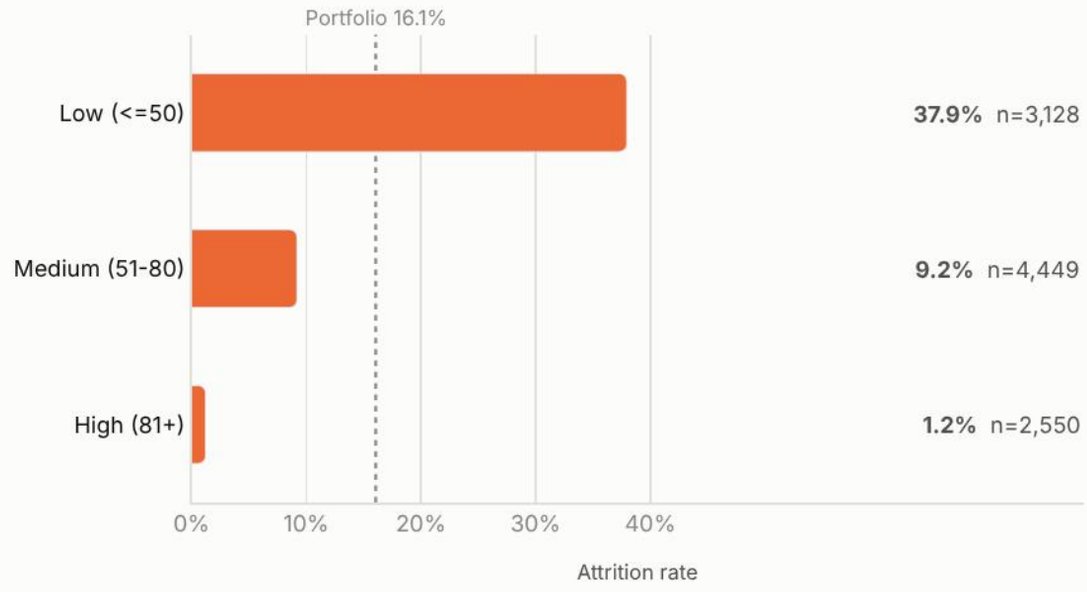
ATTRITION RATE	ACTIVE CUSTOMER RATIO	AVERAGE CREDIT LIMIT	AVERAGE UTILISATION
16.1%	**83.9%**	**$8,632**	**27.5%**
Share of customers who left	Share still with the bank	Mean limit per customer	Revolving balance / credit limit

TOTAL TRANSACTION AMOUNT	AVERAGE PRODUCTS HELD	HIGH-RISK CUSTOMERS	TRANSACTION VALUE AT RISK
$44.6M	**3.81**	**1,720**	**8.1%**
Sum over the last 12 months	Bank products per customer	17.0% of customers · 8.1% of txn value	Txn amount held by High-risk customers

Attrition rate by transaction frequency

View as table

Customers grouped by number of transactions in the last 12 months



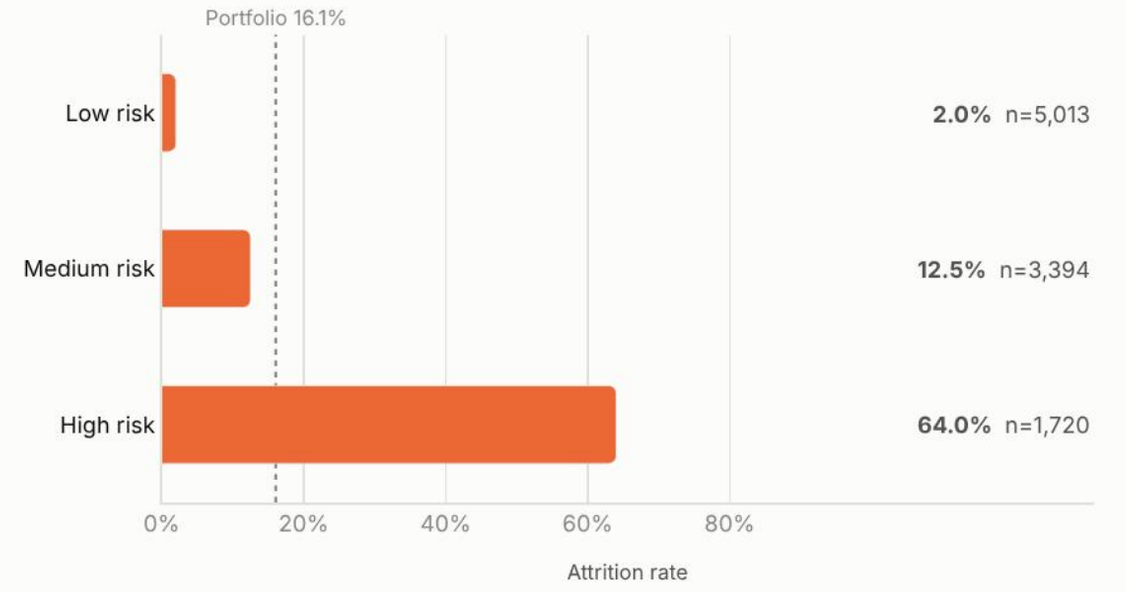
Why this chart: a short ranked bar chart compares three ordered groups against a portfolio reference line without distorting scale.

Insight (full portfolio): Customers making 50 or fewer transactions a year leave at 37.9%, about 31 times the 1.2% rate of customers with 81 or more. Transaction frequency is the strongest single warning signal in the data.

Attrition rate by retention-risk band

View as table

Validates the risk score: attrition should rise from Low to High



Why this chart: bars on a common zero baseline show whether the score truly separates customers; sample sizes are printed beside each bar.

Insight (full portfolio): The score separates customers cleanly: 2.0% attrition in Low against 64.0% in High. The High band is 17.0% of customers (1,720) yet contains 68% of all attrited customers, so it is the natural target list for retention calls.

Attrition rate by card category

Product tier from Blue to Platinum

View as table



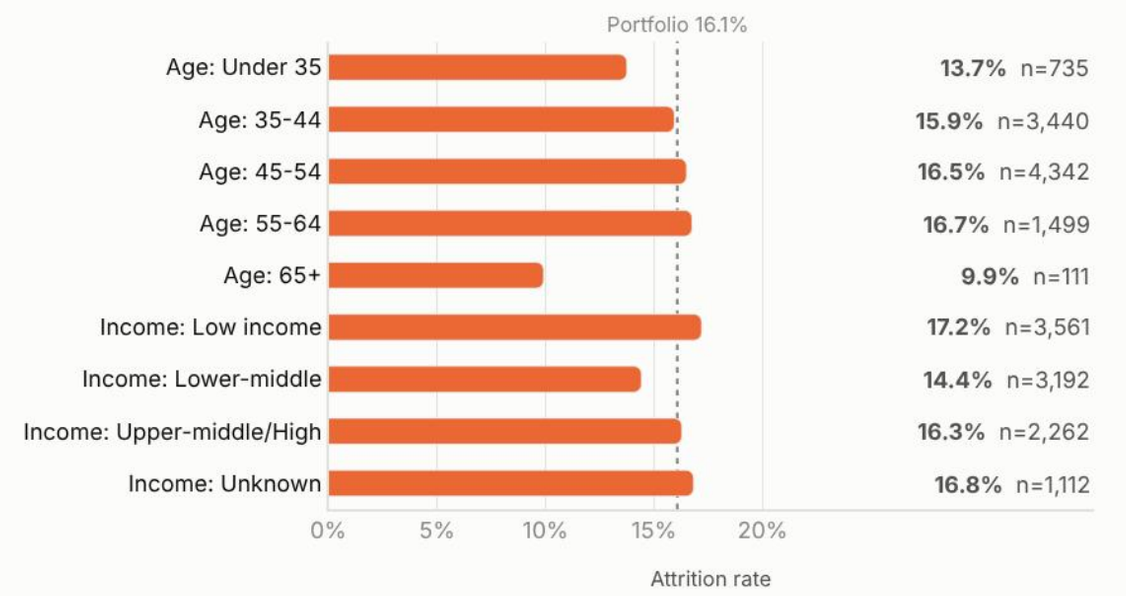
Why this chart: sample sizes are shown so a tiny tier (Platinum) is not over-read as a trend.

Insight (full portfolio): Card tier is a weak driver: Blue 16.1%, Silver 14.8%, Gold 18.1%. Platinum (25%) rests on only 20 customers, so it should not drive strategy.

Attrition rate by age band and income segment

Do customer demographics explain attrition?

View as table



Why this chart: one shared axis and the portfolio line make small differences look as small as they are.

Insight (full portfolio): Across groups with 300+ customers, attrition varies by at most 3.0 percentage points by age or income. Demographics barely explain who leaves, so retention should target behaviour, not who the customer is.

Source: Goyal, S. (2020). Credit Card Customers [Data set]. Kaggle. Filters apply to every KPI, chart and table on both pages. Insight sentences under each chart describe the full portfolio. Risk bands come from a transparent points score (see report).

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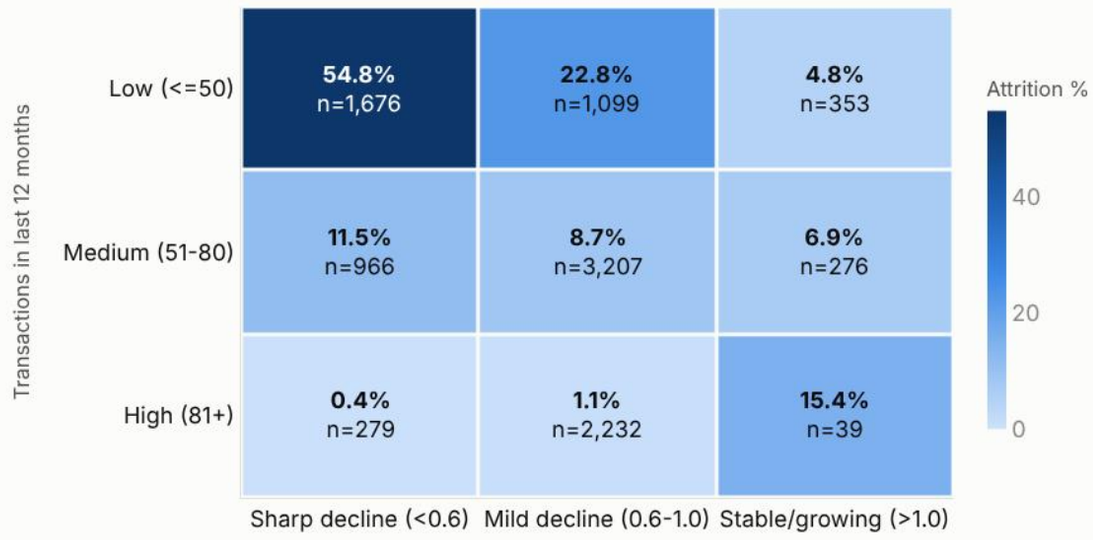
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Where attrition concentrates: activity x spending trend

View as table

Attrition rate (colour) and customers (n) for each combination



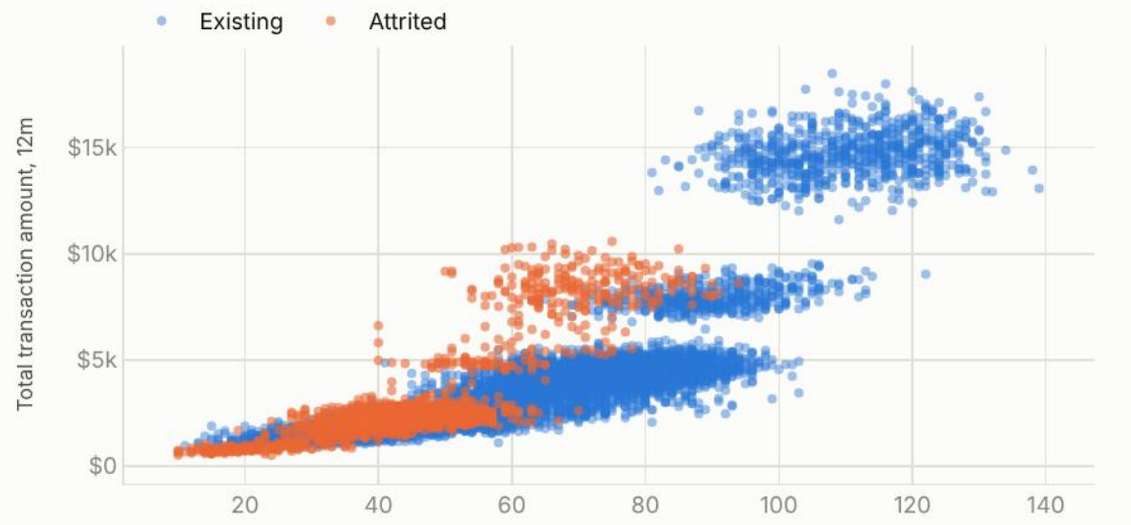
Why this chart: a heatmap suits a two-way grid of ordered categories; the numbers in each cell remove any need to judge shade.

Insight (full portfolio): Customers who are both low-activity and cutting spend churn at 54.8% (1,676 customers), and account for 56% of all attrition. Highly active customers without a sharp decline churn at 1.3%.

Transaction count against transaction amount

View as table

Each dot is one customer



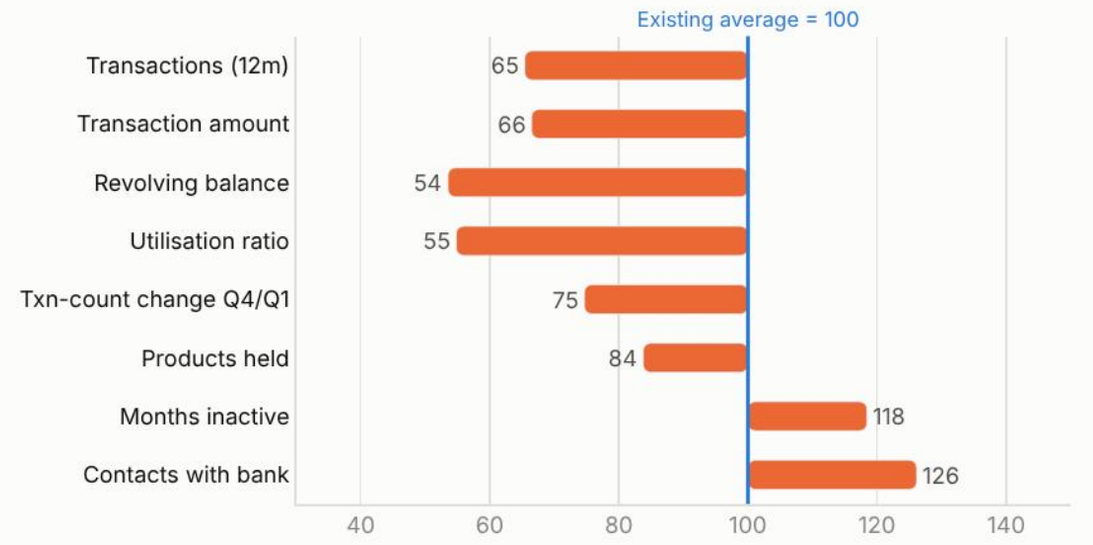
Why this chart: a scatter shows the shape of the two groups, which averages would hide. Two colours only, kept consistent across the dashboard.

Insight (full portfolio): 85% of attrited customers made 60 or fewer transactions, against 31% of existing customers. Attrited customers are already far less active than the rest, so activity-based alerts can flag accounts at risk.

How attrited customers differ from existing customers

View as table

Average of each measure, indexed so existing customers = 100



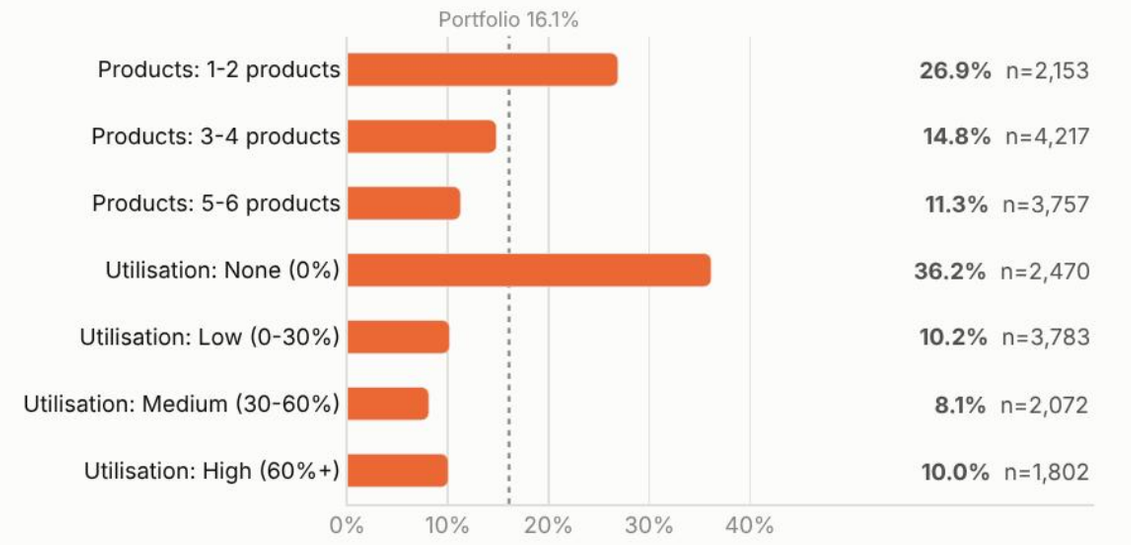
Why this chart: indexing puts measures with different units on one axis, so no dual-axis chart is needed.

Insight (full portfolio): Attrited customers transact 35% less often and hold 46% less revolving balance than existing customers, yet contact the bank 26% more. They are disengaging while contacting the bank more, which may indicate unresolved service issues (a hypothesis to test, as the data holds no complaint records).

Attrition by relationship depth and card usage

View as table

Products held and credit-utilisation band



Why this chart: the same bar form as the overview lets a reader compare engagement drivers directly.

Insight (full portfolio): Customers with 1-2 products churn at 26.9% against 11.3% with 5-6 products. Customers who never use their revolving credit churn at 36.2% against 9.6% for everyone else. Depth of relationship and card usage both protect retention.

Customer segmentation: retention-risk bands

Size, attrition and value of each band (responds to filters)

Risk band	Customers	Attrited	Attrition rate	Avg credit limit	Avg txn amount	Share of txn value
Low	5,013	101	2.0%	$8,578	$5,763	64.8%
Medium	3,394	425	12.5%	$9,111	$3,560	27.1%
High	1,720	1,101	64.0%	$7,846	$2,109	8.1%

Insight (full portfolio): High-risk customers hold only 8.1% of transaction value. Churn is concentrated in low-activity accounts, so a low-cost digital campaign suits most of them, with personal outreach reserved for the highest-value names.

Priority outreach list: highest-value high-risk customers

Top 10 customers in the High band by 12-month transaction amount (responds to filters)

Customer ID	Card	Risk score (0-12)	Model churn probability	Txns (12m)	Txn amount	Credit limit	Months inactive
711745233	Blue	7	91%	69	$9,351	$13,894	2
711161133	Blue	7	87%	59	$9,338	$6,071	3
720353808	Silver	7	92%	50	$9,177	$34,516	2
715493733	Blue	7	91%	64	$9,088	$5,735	3
714544908	Blue	7	46%	68	$8,979	$6,224	3
715150983	Blue	9	90%	60	$8,898	$26,021	3
710679933	Blue	7	95%	58	$8,660	$15,633	5
709977933	Blue	9	92%	60	$8,517	$9,959	3
712509108	Gold	9	97%	68	$8,356	$15,164	3
708821733	Silver	7	79%	70	$8,301	$19,713	3

Insight (full portfolio): These are the accounts where a personal call protects the most value. Customer IDs are shown for operational use only; this list should be access-controlled.

Source: Goyal, S. (2020). Credit Card Customers [Data set]. Kaggle. Filters apply to every KPI, chart and table on both pages. Insight sentences under each chart describe the full portfolio. Risk bands come from a transparent points score (see report).